

Q30 Ans: D

Property	α -particles (${}^4_2\text{He}$)	β -particles (${}^0_{-1}\text{e}$)	γ -rays (γ)
Stopped by	10^{-2} mm aluminium or a few cm of air	5 mm aluminum	100 mm lead

Since the radiation is not affected by the 7 mm aluminum sheet, beta radiation is absent.

Since the presence of 0.1 mm aluminum sheet decreases the count rate, alpha radiation must be present and is blocked by the sheet.

Since the 10 cm lead reduces the radiation to nearly zero, gamma radiation must be present and is blocked by the lead.

